FEAT-ZD-48600 · DES-03 Understand HLD or LLD+ Doubts Clarifications · Questions, assumptions and answers

Design agent · item feat.clarifications · source hobasa-design-agent/outputs/FEAT-ZD-48600/feat.clarifications.md · approved by Architect sim.architect

feat.clarifications: FEAT-ZD-48600

Questions, assumptions and answers for the coding lanes of FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is DES-03, loop 1. The workbook outcome: "Assumptions, Questions or Clarifications would be looped till the developer has achieved complete clarity" (record:brief).

This file reads the approved design (feat.design), stories (feat.stories) and TRD (feat.trd) the way a developer about to start would. It lists every doubt found, answers each one from an approved source where possible, states the assumption used where no source answers it, and lists what is still open with an owner. It adds no scope and changes no approved decision. Answers marked **agent judgement** are this agent's reading; no input states them as fact.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DES-03 Understand HLD or LLD + doubts clarifications (attempt 1)	record:brief
Trigger	Task creation	record:brief
Owner / gate	Architect / automated checks, independent judge, sampled human evaluation	record:brief
Inputs	feat.design (DES-02, revision 722a98f), feat.stories (REQ-08, revision 0590ea4), feat.trd (REQ-07, revision 3637e51)	record:brief; mcp:hobasa-context (context_list)
Background read	feat.screens (DES-02, approved)	journey:feat.screens
Other files read	Each repository's `ci.py` / `ci.mjs` and `CLAUDE.md`; `ap_portal/package.json`; DES-02 gate decisions	file:ap-aging-sample/ci.py; file:ap-pipeline-sample/ci.py; file:ap-portal-sample/ci.mjs; file:ap-portal-sample/package.json; file:ap-aging-sample/CLAUDE.md; file:ap-pipeline-sample/CLAUDE.md; file:ap-forecast-sample/CLAUDE.md; file:ap-portal-sample/CLAUDE.md; file:hobasa-design-agent/steps/FEAT-ZD-48600/DES-02/gate_decisions_1.json
Code read	The four repositories' source and tests from the step's tool view	tool:repo
Next step	DES-04 Task Creation reads this file	record:brief (design-steps)

0. Result of this loop

- **No blocking question.** Every story can start on the approved design. Each item still open either has an interim answer the lanes code against, or gates release rather than coding (sections 2 to 7 and 10). Agent judgement: most interim answers are confined to one function, as feat.trd IF-1.7 states for OQ-B6. The exceptions are TQ-3 / DQ-1: a transport between ap_aging and ap_portal would reopen DES-01 (feat.design A8).
- **Precedence rule used for every answer:** the story acceptance criteria and the TRD say *what* must hold. feat.design (approved at DES-02 by the Architect and the UX lead) says *how*. Where the two differ only in form (a type hint, a helper name, `.get` instead of `[]`), follow feat.design. No behavioural conflict was found between the three inputs (section 1). Agent judgement, from journey:feat.design D2 ("Conflict with an approved output: None").

- **What the DES-02 approval does and does not settle.** Both DES-02 approvals carry only the comment "automated checks and the independent judge passed". They record no answer to TQ-1 to TQ-11. The lanes therefore build on feat.design's interim answers (feat.design D3: TQ-3 → DD-3, TQ-6 → allow-list, TQ-7 → plain header row, TQ-9 → `KeyError`, TQ-10 → DD-6, TQ-11 → DD-5). Every acceptance criterion marked **[Pending <ID>]** in feat.stories keeps that marker (file:hobasa-design-agent/steps/FEAT-ZD-48600/DES-02/gate_decisions_1.json; feat.stories SQ-3).

1. Differences between the inputs, and which one to follow

ID	Topic	feat.trd / feat.stories says	feat.design says	Follow	Why	Source
X-1	`delays_by_vendor` input type	`records: List[Dict]` (IF-2; TK-02.1)	`records: Iterable[Dict]`	feat.design	A list argument behaves identically; the design notes this as the only difference	feat.trd (IF-2); feat.stories (TK-02.1); feat.design (B1.2)
X-2	Paid test in `delays_by_vendor`	`record["paid"]` is true (IF-2.2)	`record.get("paid")` is truthy	feat.design	Same result for every `to_model` record, which always has `paid`; matches the `aging_buckets` idiom. A hand-built record without `paid` is skipped, not an error	feat.trd (IF-2.2, F-2); feat.design (B1.2, NF-3); tool:repo (`transform.py`, `aging.py`)
X-3	Non-array `rows` in `renderOverdueRisk`	"missing or empty" → `''` (IF-7.1; AC-07.4)	missing, `null` or `[]` → `''`; any other non-array → `TypeError`	feat.design	Extends IF-7.1 without contradicting it; DQ-3 still open	feat.trd (IF-7.1); feat.stories (AC-07.4); feat.design (B4.2 step 1, DD-8)
X-4	Serialising the amount	`str(amount.quantize(Decimal("0.01")))` (TK-06.1; IF-6)	`str(Decimal(r["amount"]).quantize(TWO_PLACES))`	feat.design	Same result for a `Decimal` input; the design form also accepts a string amount	feat.stories (TK-06.1); feat.trd (IF-6); feat.design (B3.2)
X-5	Where the ap_portal version lives	`2.3.0` "in `index.js` and `package.json`" (TK-07.5)	`VERSION` step in `index.js` and `package.json` (B4.1)	Both files	The repository note names only `src/ap_portal/index.js`, but `package.json` also carries `"version": "2.3.0"`. Change both, or they drift apart	feat.stories (TK-07.5); feat.design (B4.1); file:ap-portal-sample/CLAUDE.md; file:ap-portal-sample/package.json
X-6	Tests for the ap_portal `TypeError` branches	Not in the TRD tests	B4.5 lists "rejects non-string amounts" only	Add tests for every branch (CL-FE6)	Coverage of at least 90% is a CI gate	feat.design (B4.4, B4.5); file:ap-portal-sample/ci.mjs

2. Cross-lane clarifications

[['ID', 'Doubt', 'Answer', 'Status', 'Source'], ['CL-X1', 'May a lane start while TQ / OQ / PQ items are open (feat.stories SQ-4)?', 'Ye

3. Data Engineering lane: ap_pipeline (STORY-01, STORY-02; DT-1, DT-2)

ID	Doubt	Answer	Status	Source
CL-DE1	Does the `paid_date` parse go inside the existing `try` block?	No. Step 1 builds the record exactly as today inside the existing `try`. Step 2 runs only for a row that passed step 1: `_parse_paid_date(r.get("paid_date"))`. On `ValueError` it appends `{"row": r, "reason": "paid_date"}` (the literal, held in `PAID_DATE_REASON`) and adds no record. If the parse sat inside the existing `try`, the reason would become `"ValueError"` and AC-01.4 would fail	Answered	feat.design (B1.2 steps 1 to 3, DD-6); feat.trd (IF-1.4, IF-1.5, F-3); feat.stories (AC-01.4)
CL-DE2	Missing column, `None`, blank or whitespace-only `paid_date`?	All give `paid_date = None`, and the row is accepted: `raw = (r.get("paid_date") or "").strip()`. A value with surrounding spaces is parsed after `strip()`	Answered	feat.trd (IF-1.1, IF-1.2, F-1); feat.design (B1.2, B1.4)
CL-DE3	Should `due_date` be stripped too, for consistency?	No. Today `due_date` is parsed without `strip()`. Changing it would change existing reject behaviour, against "every existing key and value is unchanged"	Answered	tool:repo (`transform.py`); feat.stories (AC-01.8); feat.trd (IF-1.6)
CL-DE4	Which invalid values should the tests use?	`"2026-13-01"` and `"not-a-date"` (TRD T-2). Agent judgement: avoid compact forms such as an 8-digit date without dashes. Newer Python versions of `date.fromisoformat` accept more formats than 3.9, so such a test could pass or fail depending on the CI interpreter (CL-X3)	Answered	feat.trd (T-2, IF-1.3); feat.stories (AC-01.4)
CL-DE5	Unpaid row with a valid `paid_date` (OQ-B6)	Accepted; the date is kept on the record; `delays_by_vendor` ignores it because the row is unpaid	Interim (DD-7); OQ-B6 open	feat.design (DD-7); feat.trd (IF-1.7); feat.stories (AC-01.7, AC-02.6)
CL-DE6	A row whose `paid` value is not `"true"` (for example `"yes"`) but has a `paid_date`	`paid` is `True` only for `"true"`, case-insensitive. Such a row is an unpaid row with a `paid_date`, so CL-DE5 applies	Answered (existing behaviour)	tool:repo (`transform.py`); feat.trd (F-2)

ID	Doubt	Answer	Status	Source
CL-DE7	`paid_date` before `due_date`, or after the run date	Accepted. No input makes either a reject reason. The negative `delay_days` is kept with no floor (AC-02.4). A future `paid_date` is not checked (assumption CA-3)	Answered / assumption	feat.prd via feat.trd (IF-2.3); feat.stories (AC-02.4)
CL-DE8	Row invalid in an existing field **and** in `paid_date`	Keeps today's reason (an exception type name such as `ValueError`)	Interim (DD-6); TQ-10 open	feat.design (DD-6); feat.trd (IF-1.5, T-A3); feat.stories (AC-01.6)
CL-DE9	Is `delays_by_vendor` exported from `ap_pipeline/__init__.py`?	Not in this feature: the package has no `__all__`. Import it as `from ap_pipeline.delays import delays_by_vendor`. Agent judgement: the end-to-end harness would use the same path, though neither IF-8 nor B5 states it	Interim; DQ-4 open	feat.design (DQ-4); tool:repo (`ap_pipeline/__init__.py`)
CL-DE10	Can the existing tests stay exactly as they are?	Yes. `test_run_maps_and_rejects` expects `reject_reasons["ValueError"]` for the bad `due_date` row, and that still holds. Its feed has no `paid_date` column, so it doubles as a feed v2 check	Answered	tool:repo (`tests/test_pipeline.py`); feat.design (B1.5)
CL-DE11	Feed v3 is not yet in production (PQ-6). Does that block DT-1?	No. Build and test on synthetic v3 rows; feed v2 keeps loading (IF-1.1). It gates release, not coding	Answered	feat.design (Part C "External dependencies"); feat.trd (R-4)

4. Data Science lane: ap_forecast (STORY-03, STORY-04; DT-3, DT-4)

ID	Doubt	Answer	Status	Source
CL-DS1	How do I build boundary histories that predict exactly 30 or 10?	Compare on the rounded prediction, not the raw mean. Python's `round` sends a half-way mean to the even integer. The existing test shows this: `predict_delay_days([-5, 5])` is 2 (values `[0, 5]`). Agent judgement: use single-value histories (`[30]`, `[31]`, `[10]`, `[11]`), whose prediction is the value itself. Use half-way means only in a test written for R-6 on purpose	Answered	feat.trd (IF-3.4, R-6, T-6); tool:repo (`predict.py`, `tests/test_forecast.py`); feat.design (B2.5); record:memory (L-007)

ID	Doubt	Answer	Status	Source
CL-DS2	`history` is an `Iterable`. Can it be a generator?	Yes, if `vendor_risk` calls `predict_delay_days(history)` exactly once and compares the stored result. A second pass over a generator would see it empty (agent judgement on Python iterators)	Answered	feat.trd (IF-3.1); feat.design (B2.2)
CL-DS3	Validate history values in `vendor_risk`?	No. `predict_delay_days` applies `int(d)`, and anything it cannot convert raises from there, as today	Answered	feat.design (B2.4); tool:repo (`predict.py`)
CL-DS4	Does `risk.py` touch the LLM path?	No: no LLM, no network, no read of `models.json`. Design choice: `risk.py` imports only `predict_delay_days`, not `explain`. For AC-03.7, patch `ap_forecast.explain.load_config` to fail and assert that `vendor_risk` still returns	Answered	feat.trd (IF-3.2); feat.design (B2.2, B2.5); feat.stories (AC-03.7)
CL-DS5	`build_explanation_request(..., risk='')` or a value outside high / medium / low	Interim rule: "given" means `risk is not None`, so any non-`None` value is appended as `f" Risk level: {risk}."`, with no validation. An empty string therefore yields `" Risk level: ."`. This falls under TQ-4	Interim (IF-4.5); TQ-4 open	feat.trd (IF-4.2, IF-4.5, T-A2); feat.stories (AC-04.5)
CL-DS6	Must anything new call `build_explanation_request` with `risk`?	No. IF-8 does not call it, so STORY-04 is not part of the end-to-end story. Only the parameter and its tests are in scope	Answered	feat.stories (SA-5); feat.trd (IF-8)
CL-DS7	PROV-ACME-2026-11 also changes `build_explanation_request` (PQ-5)	Build and test now. T-7 guards the `risk` parameter. Whoever merges second rebases and re-runs T-7. The order of the two changes is still open	Answered for coding; PQ-5 open	feat.design (Part C "External dependencies"); feat.trd (R-3)
CL-DS8	Sending the risk label to `acme-llm` (TQ-8)	Code it as specified; provider and model are unchanged. Confirmation of the data flow is the Architect's (TQ-8), and it gates release, not coding	Interim; TQ-8 open	feat.design (SE-3); feat.trd (S-5)
CL-DS9	Early payments only	`'low'`, because negative delays count as 0 in the prediction	Interim; TQ-5 open	feat.trd (F-6, R-5); feat.stories (AC-03.6)

5. Backend lane: ap_aging (STORY-05, STORY-06; DT-5, DT-6)

ID	Doubt	Answer	Status	Source
CL-BE1	Reuse `bucket_for` for "more than 30 days"?	No. `bucket_for(30)` returns `"31-60"`, so exactly-30 would be listed, against AC-05.2 (PRD AC-14). Use `days > OVERDUE_AFTER_DAYS` (30) directly. Leave `bucket_for` unchanged; the regression test pins `bucket_for(30) == "31-60"` as current behaviour (TQ-1 decides any fix as a separate item)	Answered	feat.design (DD-4, B3.5); feat.trd (F-10, IF-5.3, R-1); tool:repo (`aging.py`)
CL-BE2	Where does `as_of` come from? Default it to today?	No default. `as_of` is a required argument from the caller, which keeps the function pure. Tests use `AS_OF = date(2026, 10, 25)`, as the existing tests do. Agent judgement: build due dates as `AS_OF - timedelta(days=N)` so that `days` is exactly 30, 31, 42 or 75	Answered	feat.design (NF-10, B3.2, B3.5); tool:repo (`tests/test_aging.py`)
CL-BE3	Amount input types	`Decimal` from ap_pipeline; `Decimal(inv["amount"])` also accepts a string, as `aging_buckets` does. Then `quantize(Decimal("0.01"))` with the default context (round-half-even). No `float` should ever arrive; no float guard is added (assumption CA-4; CQ-6)	Answered / interim (DD-10, DQ-2)	feat.design (B3.2, DD-10); feat.trd (IF-5.5); record:memory (D-2026-044)
CL-BE4	Missing `vendor`, `id` or `due_date` on an invoice	Missing `vendor`: `KeyError`, with a test (AC-05.10). Agent judgement from the B3.2 algorithm: `due_date` is read for every unpaid invoice, so a missing one raises `KeyError`. `id` and `vendor` are read only when the row is built, so a paid or not-overdue invoice that lacks them does not raise; write the AC-05.10 test with an overdue, unpaid invoice	Interim; TQ-9 open	feat.design (DD-8, B3.4); feat.trd (IF-5.7)
CL-BE5	Sort	`rows.sort(key=lambda row: (-row["days"], row["id"]))`: days descending, then `id` ascending as text. Pipeline ids are strings, so the comparison is well defined	Answered	feat.trd (IF-5.6, F-2); feat.design (B3.2)

ID	Doubt	Answer	Status	Source
CL-BE6	Duplicate invoice ids, negative amounts (credit notes)	No input addresses either. Both are passed through: duplicates are both listed, and a negative amount serialises as `"-15.00"`-style text, which the ap_portal pattern accepts (assumption CA-5)	Assumption	feat.design (B3.2, B4.2)
CL-BE7	Non-finite amounts such as `NaN` or `Infinity`	Agent judgement on Python `Decimal`: `to_model` builds `Decimal(r["amount"])` with no finiteness check, so such a feed value could reach ap_aging. `quantize` would then return `NaN` or raise `InvalidOperation`, and ap_portal would throw `TypeError` for `"NaN"`. Agent judgement: this is existing ap_pipeline behaviour that no input names, so no lane adds handling in this feature (CA-6, CQ-1)	Assumption; CQ-1 open	tool:repo (`transform.py`); feat.design (B4.4); feat.trd (F-2)
CL-BE8	`overdue_rows_to_json` output format	`json.dumps` of a list of dicts with keys in CT-1 order: `id`, `vendor`, `days` (`int`), `amount` (two-decimal string), `risk`; `'[]'` for no rows. Agent judgement: tests should `json.loads` the result and compare values, not compare whole raw strings, because separators and ASCII escaping are `json.dumps` defaults	Answered	feat.design (A3 CT-1, B3.2); feat.trd (IF-6)
CL-BE9	Risk values from `risk_by_vendor`	Passed through as given; missing vendor gives `'unknown'`. ap_aging does not validate them, and ap_portal allow-lists the class	Answered	feat.trd (IF-5.4, IF-7.4); feat.design (DC-3)
CL-BE10	Transport from ap_aging to ap_portal	None is added. ap_aging returns the CT-1 string, and the consumer (in this feature, only the end-to-end harness) parses it. A transport would re-trigger DES-01	Interim (DD-3); TQ-3, DQ-1 open	feat.design (DD-3, A3, A8); feat.trd (R-7)

6. Frontend lane: ap_portal (STORY-07; DT-7)

ID	Doubt	Answer	Status	Source
CL-FE1	What is the exact expected output?	feat.screens FS-1: one line, `<table class="overdue-risk">`, a header row with no class (`Invoice`, `Vendor`, `Days overdue`, `Amount`), one `<tr class="risk-…">` per row with four cells, the amount cell `class="amount"`, then `</table>`. No `<thead>` / `<tbody>`, no inline style, no tokens. FS-2: `''`	Answered (TQ-7 open on header form)	journey:feat.screens (sections 2 to 4); feat.trd (IF-7.2, IF-7.3)
CL-FE2	Re-sort rows in the portal?	No. Keep the given order; ap_aging sorts	Answered	journey:feat.screens (FS-1 annotations); feat.stories (AC-07.1)
CL-FE3	Amount handling	String only, pattern `^-?\d+(\.\d{1,2})?$`; pad to two decimals as a string (`"980.5"` → `980.50`, `"980"` → `980.00`). A non-string, an exponent, a thousands separator or three decimals throws `TypeError`. Never `Number()`, `parseFloat` or `formatAmount`	Interim (DD-5, DD-8); TQ-11, DQ-3 open	feat.design (B4.2, B4.4); feat.trd (IF-7.5)
CL-FE4	`days`	Must satisfy `Number.isInteger(row.days)`, else `TypeError`; rendered as integer text	Interim (DD-8); DQ-3 open	feat.design (B4.2 step 3, B4.4)
CL-FE5	Escaping	`escapeHtml(String(value))` on `id` and `vendor` for `&`, `<`, `>`, `"`, `'`. The class comes from the allow-list `RISK_CLASSES`, so an unexpected value renders `risk-unknown`	Interim; OQ-B8, TQ-6 open	feat.design (B4.2); feat.trd (IF-7.4, IF-7.7)
CL-FE6	Which tests beyond feat.design B4.5?	Agent judgement, to keep line coverage of at least 90%: also test a non-array `rows` (for example the unparsed JSON string) → `TypeError`; a non-integer `days` → `TypeError`; `null` → `''`; an amount with an exponent, a thousands separator or three decimals → `TypeError`	Answered	feat.design (B4.4, B4.5); file:ap-portal-sample/ci.mjs
CL-FE7	Touch `formatAmount`, `LABELS` or `renderAgingSummary`?	No. They stay unchanged and their tests pass unchanged; fixing `formatAmount` is TQ-2, a separate item	Answered	feat.design (B4.1, DD-5); feat.trd (C-9, F-12)

ID	Doubt	Answer	Status	Source
CL-FE8	Show the risk as text or a column? Add row actions?	No to both. The row class is the only risk signal, and there are no row actions	Interim; OQ-B9, PQ-P7 open	feat.trd (IF-7.8); journey:feat.screens (section 7)
CL-FE9	`id` or `vendor` that is not a string (`null`, a number)	CT-1 guarantees strings. The design converts with `String(value)` and adds no check, so `null` would render as the text `null` (CQ-2)	Assumption; CQ-2 open	feat.design (A3 CT-1, B4.2)
CL-FE10	How do tests load the package?	`require('ap_portal')` with `NODE_PATH=src`, through `node:test`	Answered	tool:repo (`tests/portal.test.js`); file:ap-portal-sample/ci.mjs

7. Testing Agent: end-to-end (STORY-08)

ID	Doubt	Answer	Status	Source
CL-T1	Wiring	`records, report = pipeline.run(csv_text)`; `delays = delays_by_vendor(records)`; `risk_by_vendor = {v: vendor_risk(h) for v, h in delays.items()}`; `rows = overdue_with_risk(records, as_of, risk_by_vendor)`; `payload = overdue_rows_to_json(rows)`; Node.js 22 `renderOverdueRisk(JSON.parse(payload))`	Answered	feat.trd (IF-8); feat.design (B5)
CL-T2	`as_of`	Agent judgement: a fixed date, not today, so the exactly-30-days invoice stays at 30 on every run	Assumption	feat.stories (AC-08.1); feat.design (NF-10)
CL-T3	Feed content	It must include an exactly-30-days invoice, a paid invoice, a row with an invalid `paid_date` and a vendor with no paid history	Answered	feat.trd (T-17); feat.stories (AC-08.1, AC-08.4)
CL-T4	Who wires the stages in production?	No input names a component; in this feature only the harness wires them	Open (DQ-1)	feat.design (DQ-1, A4)
CL-T5	When does it run?	After the four lanes are merged. It is not in the coding plan	Answered	feat.design (B5)

8. Assumptions used by the lanes (new in this step)

ID	Assumption	Used in	Basis
CA-1	Where the TRD or stories and feat.design differ only in form, feat.design governs (section 1)	All lanes	feat.design approved at DES-02; no behavioural conflict found (feat.design D2)
CA-2	Interim answers in feat.design D3 hold until an owner answer is recorded; [Pending] criteria stay pending	All lanes	file:hobasa-design-agent/steps/FEAT-ZD-48600/DES-02/gate_decisions_1.json; feat.stories (SA-2, SQ-3)
CA-3	A `paid_date` after the run date is accepted (no input makes it invalid)	DE	feat.trd (IF-1.3, IF-1.4)
CA-4	No float reaches `overdue_with_risk`; no float guard is added	BE	record:memory (D-2026-044); feat.trd (F-2)
CA-5	Duplicate ids and negative amounts pass through unchanged	BE, FE	feat.design (B3.2, B4.2)
CA-6	Non-finite feed amounts are existing ap_pipeline behaviour and out of scope (agent judgement: no input names them, and the stories add no scope beyond S-1 to S-7)	BE, FE	feat.stories (header, section 8 "No new scope"); tool:repo (`transform.py`)
CA-7	Story labels from feat.stories are the IDs for branches and commits until DES-04 sets task IDs	All lanes	record:memory (conventions); file:ap-aging-sample/CLAUDE.md

9. Open questions

9.1 New in this step (all non-blocking)

ID	Question	Interim answer	Owner (proposed)	What changes if answered differently
CQ-1	Should `to_model` reject non-finite amounts (`NaN`, `Infinity`)? Today `Decimal(...)` may accept them (agent judgement on Python)	Out of scope; no handling added (CA-6)	Architect / PM (new item)	A new reject reason in ap_pipeline only
CQ-2	Should `renderOverdueRisk` throw `TypeError` for a non-string `id` or `vendor`, like the amount and `days` checks (DD-8)?	`String(value)`, no check	Architect (with DQ-3)	One check and one test in ap_portal
CQ-3	Which Python version is `/usr/bin/python3` on the CI host? Should CI add a Python 3.9 syntax gate?	Developers keep to 3.9 syntax themselves (CL-X3)	Architect / DevOps	CI script only
CQ-4	Should `build_explanation_request` restrict `risk` to `high` / `medium` / `low` (an empty string gives "`Risk level: .`")? Part of TQ-4	Append any non-`None` value	Architect	`explain.py` and T-7 only

ID	Question	Interim answer	Owner (proposed)	What changes if answered differently
CQ-5	Branch and commit IDs: the feat.stories labels, or the task IDs DES-04 will create?	feat.stories labels (CA-7)	PM (DES-04)	Branch names only
CQ-6	Should `overdue_with_risk` reject a `float` amount with `TypeError` instead of converting it?	No guard (CA-4)	Architect	One check and one test in ap_aging

9.2 Carried (unchanged; none blocks coding)

Group	IDs	Interim answer used	Source
TRD	TQ-1 to TQ-11	feat.design D3 (TQ-3 → DD-3, TQ-6 → allow-list, TQ-7 → plain header, TQ-9 → `KeyError`, TQ-10 → DD-6, TQ-11 → DD-5); TQ-1, TQ-2 are separate items; TQ-4, TQ-5, TQ-8 as in section 4	feat.trd (7.2); feat.design (D3)
Design	DQ-1 to DQ-5	As in feat.design D3	feat.design (D3)
Stories	SQ-1 to SQ-4	SQ-4 answered for coding in CL-X1, pending PM confirmation	feat.stories (6.1)
PRD / BRD / prototype	OQ-B1 to OQ-B10, PQ-1 to PQ-6, PQ-P1 to PQ-P7	As in feat.trd and feat.design	feat.stories (6.2); feat.trd (7.3)

10. Readiness per story

Story	Lane / task	Ready to build	Pending markers kept	Gates release, not coding
STORY-01	DE / DT-1	Yes	AC-01.6 (TQ-10), AC-01.7 (OQ-B6)	Feed v3 in production (PQ-6)
STORY-02	DE / DT-2 (after DT-1)	Yes	AC-02.6 (OQ-B6)	—
STORY-03	DS / DT-3	Yes	AC-03.5 (PQ-1), AC-03.6 (TQ-5)	—
STORY-04	DS / DT-4	Yes	AC-04.5 (TQ-4)	PROV-ACME-2026-11 order (PQ-5); TQ-8
STORY-05	BE / DT-5	Yes	AC-05.6 (PQ-2), AC-05.10 (TQ-9)	—
STORY-06	BE / DT-6 (after DT-5)	Yes	AC-06.1 to AC-06.4 (TQ-3)	—
STORY-07	FE / DT-7	Yes	AC-07.2 (TQ-7), AC-07.6 (TQ-11, OQ-B10), AC-07.7 (OQ-B8), AC-07.8 (TQ-6), AC-07.9 (OQ-B9, PQ-P7)	—
STORY-08	Testing Agent, after merge	Yes, after the four lanes	AC-08.1 (PQ-2)	— (DQ-1 stays open, non-blocking)

Sources: feat.stories (sections 2 to 4); feat.design (Part C, D3).

11. How the loop continues

The feature issue takes answers in the form `/answer <STEP>: <text>` (file:hobasa-agentic-sdlc/features/FEAT-ZD-48600/COMMUNICATION.md). Proposed loop (agent judgement): a developer raises a new doubt there for DES-03. The next loop of this step adds it here with an ID (CL-… or CQ-…), an answer from an approved source or an interim assumption, and an owner. When an owner answers a TQ, DQ, CQ, OQ or PQ item, the matching [Pending] criteria in feat.stories (SA-2) can be updated. DES-04 reads this file as an input (design-steps).

12. Validation check

Check	Result	Source
Every input read: feat.design, feat.stories, feat.trd	Pass	record:brief
Every story and lane has its doubts listed and answered or assigned	Pass (sections 3 to 7, 10)	feat.stories (section 1)
No approved decision contradicted; differences resolved by a stated rule	Pass (section 1, CA-1)	feat.design (D2)
No new scope, technology or contract	Pass	feat.design (DD-1)
Decisions D-2026-031, D-2026-044, D-2026-050; lessons L-007, L-011	Consistent (CL-X10, CL-BE1, CL-BE3, CL-DS1)	record:memory
Blocking questions	None	—